Dayi Ethan Dong

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EDUCATION

University of California Berkeley, Berkeley, CA August, 2024 – Present
Ph.D., Mechanical Engineering
- Advisor: Dr. Negar Mehr
- Affiliated with Berkeley Artificial Intelligence Research (BAIR)
- My research focuses on multi-robot collaboration primarily in decentralized settings using generative modeling.

Yale University, New Haven, CT August, 2020 – May, 2024
B.S., Major(s): Engineering Sciences (Mechanical) – with Distinction, Computer Science – with Distinction
- Magna Cum Laude
- Cumulative GPA: 3.95; Major GPA: 3.99
- Relevant Coursework: Robot Learning; Introduction to Embedded Robotic Systems; Introduction to Robotics, Control, and Learning; Automated Decision Systems; Intelligent Robotics; Applied Planning and Optimization

TECHNICAL SKILLS

- Programming Languages: Python, C, Java, ROS, ROS2
- Software: Solidworks, Fusion 360
- Languages: English (Native), Mandarin (Fluent), Spanish (Basic)

RESEARCH EXPERIENCE

Honda Research Institute, Mountain View, CA Sep, 2025 – Dec, 2025
Robotics Software Engineering Intern
- Worked on developing improvements to autonomous driving algorithms under Dr. David Isele

Yale University, *Intelligent Autonomy Lab*, New Haven, CT November, 2021 – May, 2024
Undergraduate Research Assistant
- Conducted research on multi-agent robotic experiments and validating algorithms for search and exploration under Dr. Ian Abraham
- Collaborated with undergraduate students, graduate students, and other senior researchers on work related to ergodic search

Carnegie Mellon University, *Biorobotics Lab*, Pittsburgh, PA May, 2023 – August, 2023
Undergraduate Research Assistant
- Carnegie Mellon University's Robotics Institute for Summer Scholars (RISS) program
- Researched ergodic search and modular robotics under Dr. Howie Choset

PUBLICATIONS AND PRESENTATIONS

- **D. Dong**, M. Bhatt, A. Shrivastava, L. Peters, N. Mehr, "Residual Denoising Enables Sample-Efficient Multi-Agent Coordination on Demand," pending review at 2027 *International Conference on Robotics and Automation (ICRA),* http://arxiv.org/abs/2609.32129

- **D. Dong***, M. Bhatt*, S. Choi, N. Mehr, "MIMIC-D: Multi-modal Imitation for MultI-agent Coordination with Decentralized Diffusion Policies," in 2026 *International Conference on Robotics and Automation (ICRA),* https://arxiv.org/abs/2509.14159
    - Publication/Poster presentation

- K. Nagpal, **D. Dong**, J.B. Bouvier, N. Mehr, "Leveraging Large Language Models for Effective and Explainable Multi-Agent Credit Assignment," in 2025 *Autonomous Agents and Multiagent Systems (AAMAS),* https://arxiv.org/abs/2502.16863
    - Publication

- **D. Dong**, A. Xu, G. Gutow, H. Choset I. Abraham, "Ergodic Exploration over Meshable Surfaces," in 2025 *International Conference on Robotics and Automation (ICRA),* https://arxiv.org/abs/2503.05026
    - Publication/Poster presentation/Oral presentation

- **D. Dong**, H. Berger, I. Abraham, "Time Optimal Ergodic Search: Multiscale Coverage in Minimum Time," in *International Journal of Robotics Research*, doi.org/10.1177/02783649241273597
    - Publication/Journal extension

- **D. Dong**, H. Berger, I. Abraham, "Time Optimal Ergodic Search," in *2023 Robotics: Science and Systems (RSS)*, arxiv.org/abs/2305.11643.
    - **Best Paper Award**
    - Publication/Poster presentation/Spotlight talk

- C. Lerch, **D. Dong**, I. Abraham, "Safety-Critical Ergodic Exploration in Cluttered Environments via Control Barrier Functions," in *2023 International Conference on Robotics and Automation (ICRA),* arxiv.org/abs/2211.04310.
    - Publication/Poster presentation

LEADERSHIP/WORK EXPERIENCE

FIRST Robotics Competition Team 5419 Berkelium, *Mentor,* Berkeley, CA September, 2024 – Present
- Worked with the high school robotics team to help them throughout their competition and off season

IEEE Robotics and Automation Society, *Student Representative,* Virtual April, 2024 – Present
- Multi-Robot System Technical Committee
- Coordinated with the Chair and Co-Chairs to organize committee events like meetings, workshops
- Facilitated communication between members and students around the world
- Represented the student perspective within the committee

HONORS AND AWARDS

- Best Paper Award (Robotics: Science and Systems 2023)
- 2024 Honorable Mention for Computing Research Association's Outstanding Undergraduate Researcher Award
- Donald Warren McCroskey Memorial Prize
    - Awarded to a senior who is deserving of the greatest distinction for scholarly achievement in fields related to mechanics and its applications

GRANTS AND FELLOWSHIPS

- National Defense Science and Engineering Graduate (NDSEG) Fellowship Award [Offered]
- Berkeley Fellowship
- Dean's Research Fellowship
- Yale Summer Experience Award

SOCIETIES

- Phi Beta Kappa Honor Society
- Tau Beta Pi Engineering Honor Society
- IEEE Membership
- IEEE Robotics and Automation Society Membership
- ASME Membership